

APPLIED RESEARCH

LoadSeer: Exploiting Tensor Graph Convolutional Network for Power Load Forecasting With Spatio-Temporal Characteristics

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ABSTRACT Power load forecasting plays a crucial role in ensuring the stable operation of the power system and avoiding system collapse or resource waste caused by power shortages or surpluses. However, the complex spatio-temporal property of power load makes it difficult to predict, which poses a great challenge to the power system. Existing spatio-temporal prediction methods can only handle one factor in each dimension of time and space. In reality, power load is influenced by various factors. Especially in terms of time dimension, crowd flow, weather, and historical load all have significant impacts on load forecasting. Inspired by tensor time series, considering the structure of spatial geographic location, we propose Tensor Graph Convolutional Network for Power Load Forecasting, LoadSeer. A distance adjacency matrix is designed to represent the geographical location relationship and land use nature. A spatio-temporal processing layer integrating graph convolution module (GCN) and T-Transformer is mapped out to extract the spatio-temporal features, which are then sent to the fully connected layer to provide the refined expression. The experimental results on three public datasets, PeMSD4, PeMSD7, and PeMSD8 show that our proposed method outperforms baseline models on all indicators. To further validate the effectiveness of our proposed approach, we apply LoadSeer to real load data during the Asian Games in a certain city, and the results also demonstrate the superiority of our method.

INDEX TERMS Tensor time series, graph convolutional network, spatio-temporal sequence, electricity load forecasting.

I. INTRODUCTION

Spatio-temporal data receives increasing attention, due to the widespread application of new-generation information technology, which has become an important foundation for the construction of new digital infrastructure and has strategic value in promoting the new development of the industry. Currently, some spatio-temporal prediction models have been proposed and widely applied in various fields and tasks.

Existing research on spatio-temporal sequence prediction primarily focuses on traffic prediction. Early studies mainly emphasize time series data while neglecting

spatial correlation, leading to inaccurate prediction results, especially in cities or regions with complex road network structures. Therefore, researchers have explored how to effectively model the spatial characteristics of traffic flow by integrating both temporal and spatial features to improve the accuracy of traffic flow prediction. Voort et al. [1] Enhanced the accuracy and efficiency of traffic flow forecasting by using Kohonen maps, a self-organizing mapping neural network, in conjunction with the ARIMA time series model. Ma et al. [2] adopt a two-dimensional spatio-temporal matrix to transform the dynamic traffic data into images that depict the spatio-temporal relationships of traffic flow. They apply CNN to these image data. However, CNN is primarily suitable for processing structured data, and the topological

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structure of roads and the unstructured nature of traffic flow may result in the loss of some valuable information when using CNN to extract spatial features of traffic flow. With the rise of graph neural networks, researchers have introduced a series of models based on graph neural networks to address extracting features from unstructured information. Graph convolutional neural networks have been applied in traffic prediction and have achieved significant results.

By combining graph neural networks (GNNs) with recurrent neural networks (RNNs) to construct spatio-temporal sequence prediction models, we can effectively capture spatial and temporal features of geographical topological graphs. This approach addresses modeling geographical location networks, learning flow data propagation patterns, and adapting to dynamic changes. Lee et al. [3] propose a richer multi-graph convolutional framework, which is able to capture complex dynamic properties in road networks, such as non-Euclidean space and directional features, enabling it to perform well in urban roads. Bai et al. [4] by combining GCN with GRU, the modeling of time series and road network topology is achieved. And the weights are dynamically adjusted according to the importance of different historical time points, so as to capture the global traffic change trend more accurately. Wu et al. [5] design the Multi-Temporal Graph Neural Network (MTGNN) framework tailored for modeling multi-dimensional time series data. Feng et al. [6] propose a novel deep learning approach based on multi-component spatio-temporal graph convolutional network for traffic flow prediction. Zhao et al. [7] introduces a comprehensive prediction method combining graph convolutional networks (GCN) and Gated Recurrent Units (GRU). They use GCN to learn complex topological structures to capture spatial dependencies while employing GRU to capture dynamic changes in traffic data, effectively capturing temporal dependencies. On the other hand, Gu et al. [8] propose an entropy-based grey relational analysis method to extract spatial dependencies in traffic flow. They also utilize a two-layer recurrent neural network to predict lane-level speeds rapidly. Du et al. [9] introduce Astrologer, a graph neural Hawkes process that integrates spatial information and influences between different events to capture event propagation from historical events on graphs. The fundamental idea of Astrologer is to combine the conditional intensity function of the Hawkes process with graph convolutional recursive neural networks. Subsequently, Du et al. [10] introduce a model named Nostradamus, which captures the propagation of interesting events from historical events with spatio-temporal features using a hyperbolic graph neural Hawkes process. The core concept of Nostradamus combines the conditional intensity function of the Hawkes process with hyperbolic graph convolutional recursive neural networks. The above methods allow for a comprehensive consideration of spatial and temporal features, thereby enhancing the accuracy and efficiency of traffic flow prediction.

Similar to traffic flow prediction, electricity load forecasting also needs to consider spatio-temporal sequence characteristics. As is known, power load forecasting is related to many factors, involving climate, population mobility, special events, and historical loads. Above spatio-temporal sequence prediction models are not applicable to power load forecasting as it requires the use of multi-dimensional time series data, while the existing spatio-temporal forecasting methods can only handle one-dimensional time series data and one-dimensional spatial series data. In addition, previous frameworks often rely on given spatial adjacency graph and complex mechanism to capture spatial and temporal correlations. However, limited representations may hinder the model's ability to learn effective spatio-temporal dependencies when incomplete neighboring connections are present in the given spatial graph structure. Therefore, to improve the generalization ability and adaptability, as well as effectively handle multiple time series data, we propose a novel spatio-temporal sequence prediction model LoadSeer.

Tensor time series are advanced, where Transformer is employed to deal with the multiple time series. LoadSeer segments and vectorizes the tensor time series data, using the vectorized sequences as inputs to Transformer. Additionally, it integrates graph convolutional neural networks to extract spatial geographic distribution features. Subsequently, further extracts and combines multiple temporal and spatial evolution features through a fully connected layer, culminating in predictions at the output layer. To evaluate the performance of our proposed method, we compare it with several well-known spatio-temporal prediction algorithms. Results demonstrate a significant enhancement in prediction accuracy using LoadSeer compared to the benchmark algorithms. To further validate its practicality in real-world scenarios, we apply LoadSeer to load forecasting in a certain area during the Asian Games. The experimental results not only reflect the effectiveness of the proposed algorithm but also verify the impact of crowd flow on load.

The main contributions of this paper are as follows:

- We have discovered and verified for the first time the impact of population mobility on load forecasting through experiments. Besides, we establish a multi-dimension spatio-temporal sequence data, including weather, human traffic, history load, and geographical location information.
- Compared with existing spatio-temporal prediction methods that can only manage one-dimensional time sequence and one-dimensional space sequence, we put forward tensor-based time series, introducing multiple time series data into spatio-temporal prediction.
- We not only consider geographical location information in spatio-temporal load forecasting, but also differentiate land use attributes for the first time, and a distance adjacency matrix is defined to represent the structure. We propose LoadSeer, a spatio-temporal sequence prediction model, to comprehensively consider multiple

temporal time series influencing factors, as well as spatial geographic relation and land use properties, thereby enhancing prediction accuracy and robustness.

II. RELATED WORK

A. SPATIO-TEMPORAL SEQUENCE

Existing methods for modeling spatio-temporal dependencies can be categorized into three types: spatio-temporal statistics, artificial intelligence (AI), and physical models. Spatio-temporal statistics rely on established statistical theories, such as probability theory, including geographical weighted regression (GWR) [11], spatio-temporal autoregressive integrated moving average (ST-ARIMA) [12], spatio-temporal kriging (ST-kriging) [13], and Bayesian maximum entropy (BME) [14]. Emerging AI methods, such as machine learning and deep learning, approach spatio-temporal dependencies from a data-driven perspective [15]. These AI models have outperformed traditional statistical models in areas like weather forecasting, traffic prediction, video prediction, and energy estimation [16], [17], [18], [19], [20]. Physical models, grounded in physical mechanisms, offer plausible explanations for forecasted results, but their accuracy can not match that of statistical and AI models [16], [21], [22], [23], [24]. Combining AI and physical models can leverage their respective strengths, enhancing simulation and forecasting accuracy [25], [26].

B. POWER LOAD FORECASTING

Power load forecasting is crucial for the intelligent management of power grids. It plays a significant role in formulating power dispatch plans and enhancing the security and stability of the grid [27], [28]. In recent years, deep learning models and hybrid approaches have been extensively applied in short-term load forecasting due to their ability to extract abstract features from vast datasets. Notable models such as one-dimensional Convolutional Neural Networks (CNN) [29], Gated Recurrent Units (GRU) [30], and Long Short-Term Memory (LSTM) networks [31] have demonstrated substantial success in this domain. Additionally, the Temporal Convolution Network (TCN), which leverages multi-layer causal extended convolution and residual structures to capture dependencies across different time steps [32], has shown superior performance compared to LSTM and GRU [33]. To address the limitations of a single algorithm, combined models have emerged [34]. For example, a model integrating GRU with a dual attention mechanism enhances long-term prediction stability by accurately selecting key information through the attention block [35]. The CNN-BiLSTM-Attention model, combining CNN, bidirectional LSTM (BiLSTM), and an attention mechanism, leverages data with similar characteristics for training, thus learning the periodicity and temporal dependencies of the load, which reduces prediction errors [36]. Another study utilizes XGBoost and TCN to capture trend characteristics and nonlinear fluctuations in load series, improving forecasting

performance [37]. Furthermore, the TCN-GRU, designed for load feature extraction and prediction, has demonstrated the highest prediction accuracy compared to GRU, LSTM, and TCN models [38].

Based on the above works, this paper considers the impact of different factors (population movements, weather, and historical load) on power load forecasting, and proposes a novel tensor graph convolutional network for multi-dimension spatio-temporal sequence, LoadSeer. Combining the structure of time series and spatial geographical locations, we analyze the load of different land use properties in the surrounding area.

III. MODEL DESIGN

The overall framework of LoadSeer is illustrated in Figure 1. LoadSeer comprises three main components: the input layer, the spatio-temporal processing layer, and the output layer. The input layer receives tensor time series data X (such as weather, foot traffic, load, etc.) along with the distance adjacency matrix A , representing the geographical relationships (such as proximity to commercial areas, residential zones, hotels, etc.). In the spatio-temporal processing layer, leveraging the excellent performance of Graph Convolutional Network (GCN), we extract spatial features of regions through graph convolutional operations. Subsequently, the spatial features are concatenated with multi-dimension temporal data, then are both fed into a T-Transformer, which captures the spatio-temporal features. The results are transmitted to the fully connected layer, where the spatio-temporal features are fitted into a probability matrix.

A. THE INPUT LAYER

Tensor time series is proposed to capture coordinated changes among multiple time series, which is highly valuable for time series prediction and analyzing various variables' correlations.

Tensor time series is an $n+1$ mode tensor, denoted as $\chi \in R^{I_1 I_2 \cdots I_n T}$. At time t along the temporal dimension, the sliced tensor $\chi_t \in R^{I_1 I_2 \cdots I_n}$ is an n th-order tensor containing $I_1 I_2 \cdots I_n$ data points in the time series. Predicting the tensor time series χ_t involves using historical sliced data $\chi_1, \chi_2, \cdots, \chi_t$ to forecast future data at time $\chi_{T+\beta}$:

$$\chi_{T+\beta} = f(\chi_1, \chi_2, \cdots, \chi_t), \quad (1)$$

where f is the predictive function to be trained, β represents the prediction horizon, and $\chi_{T+\beta}$ denotes the forecast of the tensor slice at time $T + \beta$.

Given the adjacency weight matrix $A \in \mathbb{R}^{n \times n}$, where A_{ij} represents the edge weight between node v_i and node v_j . The Laplacian matrix $L = D - A$ is obtained, where $D = \text{diag}\{d_i\}$ is the degree matrix, $d_i = \sum_j A_{ij}$ is a further normalization of the Laplacian matrix, and then yields $L_s = I_N - D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$, where $I_N \in \mathbb{R}^{n \times n}$ is the identity matrix. Performing the feature decomposition on the normalized Laplacian matrix $L_s = U_s \Lambda_s U_s^T$ yields the

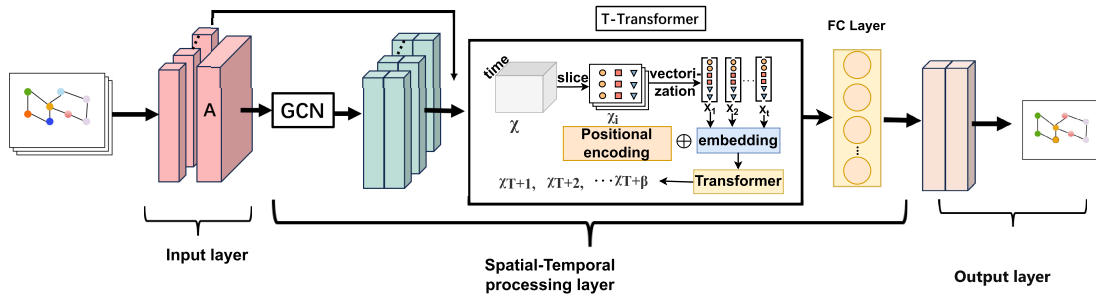


FIGURE 1. The overall framework of LoadSeer.

featurevalue matrix $\Lambda_s = \text{diag}\{\lambda_i\}$ ($\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$) and the corresponding featurevector matrix U_s .

B. THE SPATIO-TEMPORAL PROCESSING LAYER

1) GCN

The GCN module takes tensor time series data X and the obtained adjacency matrix A as the input, and extracts spatial features through graph convolution operations. Graph convolution is defined as the filter $g_\theta = \text{diag}(\theta)$ multiplied with the graph signal $x \in \mathbb{R}_n$ in the frequency domain, as shown in Equation 2:

$$g_\theta \hat{a} x = U_s g_\theta U_s^T x, \quad (2)$$

where $\hat{a}$ denotes the graph convolution operation. However, as the data dimension increases, the computational complexity of Equation 2 also significantly increases. To mitigate this issue and reduce computational complexity, a second-order Chebyshev polynomial is employed to approximate g_θ , resulting in the approximate Equation 3:

$$g_\theta \hat{a} x \approx c_0 x - c_1 D^{-\frac{1}{2}} A D^{-\frac{1}{2}} x = c \left(I_N + D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \right) x, \quad (3)$$

where $c = c_0 = -c_1, c_0$ and $-c_1$ represent the coefficients of the Chebyshev polynomial. Furthermore, by normalizing $I_N + D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$ as $\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}$, the definition of the graph convolutional layer is given by the following Equation 4:

$$h = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} x w^T, \quad (4)$$

where $\tilde{A} = A + I_N$ represents the sum of the adjacency matrix A and the identity matrix I_N , and $\tilde{D}$ represents the degree matrix of $\tilde{A}$. Furthermore, w_T represents the weight vector. According to Equation 4, the forward propagation formula for the GCN module is as follows:

$$S_t = f_{GCN}(x_t, \tilde{A}_s) = \tilde{A}_s (\tilde{A}_s x w_0^T) W_1, \quad (5)$$

where w_0^T and W_1 represent the weight vectors of the first and second convolutional layers, respectively. $S_t \in \mathbb{R}^{n \times 2}$ stands for spatial features of the graph signal at time t .

2) CONCATENATION

The spatial features from the distance adjacency matrix A of the graph model defined above, maintain spatial relationships among different regions, including relative positions and connectivity among regions such as commercial areas, residential areas, and hotel areas. Simultaneously, the time series data passes through sequence padding, normalization, and feature extraction to ensure data consistency and comparability. Then the temporal features are aligned with the spatial features for subsequent concatenation operations.

3) T-TRANSFORMER

Traditional Transformer typically accepts vectors as input. However, when dealing with multi-dimensional tensor time series data, the tensor time series need to be sliced and vectorized. The encoded vector sequence is then fed into Transformer. This operation transforms the multi-dimensional features of the tensor time series into multiple vector sequences for Transformer, thereby combining the vectorization of tensor structures with Transformer within a unified framework. We name it T-Transformer, multi-dimensional analysis, and prediction problems.

Inspired by Kiers' Nth-order tensor matrixization method proposed in 2000 [39], we slice the tensor by the tensor module k-slicing the tensor is sliced using the tensor mode k-slicing. That is, the tensor is sliced along the time dimension to obtain tensor slices for each time point. The vectorization of the tensor involves column vectorization and row vectorization. Column vectorization rearranges an Nth-order tensor $\chi \in \mathbb{R}^{I_1 I_2 \dots I_n}$ into a column vector of size $(I_1 I_2 \dots I_n) 1$, commonly denoted as $X^{(I_1 \times I_2 \times \dots \times I_n) \times 1} = \text{vec}(\chi)$. Similarly, row vectorization rearranges the tensor into a row vector.

In the T-Transformer, an embedding operation is performed on the vectorized tensor time series, transforming it into high-dimensional embedding vectors. However, these embedding vectors do not inherently contain position information about the elements in the sequence. Position encoding can preserve the position relationships of the sequence elements, thereby enhancing the expressiveness and performance of the model. We can use the most common approach, sine and cosine functions, to encode the positions.

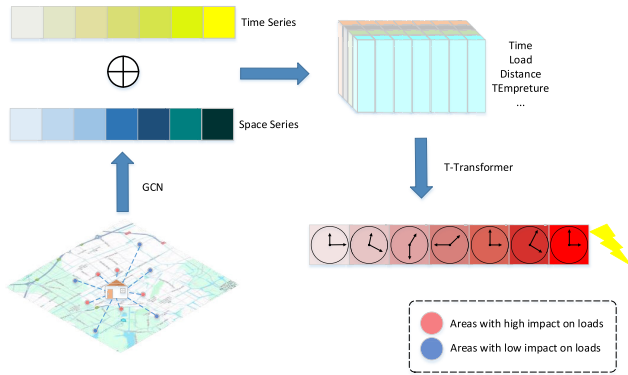


FIGURE 2. Spatio-temporal data processing.

By involving position encoding in the embedding vectors, new vectors containing position information are obtained. These new vectors serve as the input to T-Transformer for further tasks. The dimension of position encoding is usually matched with the dimension of embedding vectors to ensure additivity.

C. THE OUTPUT LAYER

The output layer is to fit the spatio-temporal features obtained from the spatio-temporal processing layer into a probability matrix and outputs the final prediction results. This probability matrix reflects the probability distribution of different prediction targets under different spatio-temporal conditions. The main task of the output layer is to generate the final prediction results based on this probability matrix. Figure 2 shows the overall data processing flow.

IV. EXPERIMENTAL EVALUATION

To validate the effectiveness of the proposed model, we first conduct extensive experiments on several commonly used public datasets and select multiple evaluation metrics to observe the results. To further verify its applicability for power load forecasting during specific events, we establish a comprehensive spatio-temporal dataset that includes historical load data, crowd flow, weather conditions, geographical relationships, and land use properties of different regions within a certain area during the Asian Games. In this section, we introduce the datasets, baselines, and evaluation metrics. Next, the experimental results are provided and analyzed.

A. DATASETS

There are very few public datasets for electricity spatio-temporal load forecasting, so we choose traffic forecasting datasets that are similar to load forecasting for the experiments. The PeMSD4, PeMSD7, and PeMSD8 datasets are commonly used for the evaluation of sequence prediction methods. These datasets were collected every 30 seconds by the California Department of Transportation's Performance Measurement System (PeMS) [40].

PeMSD4: This dataset is derived from traffic data in the San Francisco Bay Area, with 307 sensors on 29 roadways.

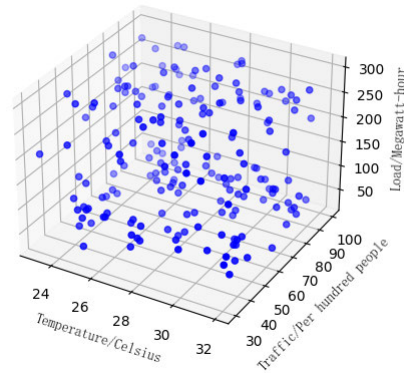


FIGURE 3. Multi-dimensional Spatio-temporal sequence data.

The dataset covers the period from January to February 2018. The first 47 days are used for training, while the remaining days are used for validation and testing.

PeMSD7: This dataset consists of traffic data from 7 districts in California, with 228 sensors. The data covers weekdays from May to June 2012. The first 34 days serve as the training set, and the remaining days are allocated for validation and testing.

PeMSD8: This dataset includes traffic data from San Bernardino, collected between July and August 2016. There are 170 detectors placed on 8 roadways. The first 50 days are used for training, and the remaining time is used for validation and testing.

Since these datasets only provide one-dimensional time-series data, we first examine the effect of the proposed method on one-dimensional time-series forecasting.

To further evaluate LoadSeer on multi-dimensional spatio-temporal sequence data, as shown in Figure 3, we establish a real-world electricity load dataset combined with synchronized weather and crowd movement, as well as geographical location relationships and land use properties around the venues during the Asian Games. The land is marked as residential, commercial, and hotel areas. This dataset has been cleaned and preprocessed by removing outliers, filling in missing data, and standardizing the values. We refer to this dataset as Flowload. In this dataset, the load is the dependent variable, while all other sequence data are treated as independent variables.

B. DATA PREPARATION

For the above public datasets, in order to divide the data at a fine-grained level, we follow the method of LSGCN [41] LSGCN[15] to preprocess. We set the standard time interval to 5 minutes, and fill in the missing values using linear interpolation, and normalize the input data using Z-score.

C. BASELINES

In order to evaluate the performance of our proposed LoadSeer, we compare it with state-of-the-art spatio-temporal forecasting models (ARIMA, LSGCN, AGCRN, and ASTGNN).

ARIMA [42]: introducing seasonal components, the cyclical and trend changes in traffic flow. The model parameters are estimated using the maximum likelihood estimation method and validated with actual traffic flow data.

LSGCN [41]: a novel graph attention network called 'cosAtt' integrating the graph convolutional network (GCN), and forming a powerful spatial gating block. By leveraging this spatial gating block and Gated Linear Unit convolutions (GLU), LSGCN efficiently captures complex spatio-temporal features, resulting in reliable prediction results. This innovative combination has significant potential in the field of spatio-temporal feature analysis.

AGCRN [43]: focusing on learning specific patterns of individual nodes in traffic prediction tasks and proposes two adaptive modules: the node-adaptive parameter learning module and the data-adaptive graph generation module, which aim to enhance the capabilities of graph convolutional networks to automatically capture fine-grained spatio-temporal dependencies in the traffic network.

ASTGNN [44]: a novel self-attention mechanism is specifically designed for transforming numerical sequence representations in the temporal dimension. This self-attention mechanism leverages local context to better capture the temporal dynamics of traffic data.

D. EVALUATION METRICS

Commonly used metrics, such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), are used for comparative experiments. RMSE is a widely used metric to measure the difference between predicted values and actual observations. It is calculated as the mean of the squared differences between the predicted and actual values and then taking the square root of the result. A lower RMSE indicates better prediction accuracy for the model. MAE is another frequently used evaluation metric that calculates the mean of the absolute differences between predicted and actual values. Unlike RMSE, MAE does not consider the squared errors, thus focusing more on the absolute differences between predicted and actual values. MAPE is an evaluation metric expressed as a percentage. It calculates the mean of the absolute percentage errors between predicted and actual values. MAPE provides insights into the relative error magnitude of a model at different load levels, which is crucial for comparing the performance of different models.

Considering these three evaluation metrics, the accuracy and reliability of the models in spatio-temporal forecasting can be comprehensively assessed. Lower RMSE and MAE values, as well as lower MAPE values, indicate better model performance.

E. EXPERIMENTAL RESULTS

LoadSeer is first compared with four benchmark methods (ARIMA, LSGCN, AGCRN, and ASTGNN) across three datasets: PeMSD4, PeMSD7, and PeMSD8. The prediction

horizon is gradually extended, and metrics RMSE, MAE, and MAPE are collectively used to gauge the predictive accuracy of the models.

As shown in Table 1, it can be easily observed that on the three datasets—PeMSD4, PeMSD7, and PeMSD8—LoadSeer shows competitive performance in MAE, MAPE, and RMSE metrics. Particularly notable is that in most cases, LoadSeer achieves SOTA results on all three datasets (except for MAPE at 15 minutes on the PeMSD4 dataset, MAE at 15 minutes on the PeMSD8 dataset, and three cases on PeMSD7), which indicates that our proposed model has a significant advantage in terms of all the indicators. From the experiments, our model demonstrates the ability to effectively control large errors and reduce the negative impact of extreme values on the overall prediction accuracy, making it perform exceptionally well in capturing the overall trend and change pattern of the data, thus improving the stability and reliability of the prediction results.

It is specifically worth noting that on the PeMSD4 dataset, LoadSeer reduces the RMSE metric to 2.91 in 60-minute forecasting, which is 1.92 lower than the previous optimal value. On the PeMSD7 dataset, LoadSeer reduces the MAPE to 9.14%. From Table 1, it is evident that LoadSeer achieves almost optimal results in the MAPE metric on all three datasets, reflecting its reliability in handling data of different magnitudes as well as excellent processing logic for both large and small data values. LSGCN is also competitive in ultra-short-term prediction due to its LSTM module's ability to efficiently capture short-term dependencies and detail changes through its recursive and gating mechanisms, while our designed T-Transformer is more inclined to capture global dependencies, which explains LoadSeer's excellent performance in slightly long-term prediction.

To further validate the effectiveness of LoadSeer on spatio-temporal load forecasting, experiments are conducted on real data during the Asian Games in a certain region. As shown in Table 2, LoadSeer exhibits a notable advantage in terms of RMSE, showcasing its superior overall predictive capability. ASTGNN achieves the best result in MAPE, with LoadSeer closely following and a difference of only 0.01.

Nonetheless, the prediction performance of LoadSeer on real dataset Flowload is very close to the true values. Figure 4 displays the comparative results between the model predictions and the actual values.

From Figure 4, it is evident that the trend of predicted values aligns closely with the true values, and the maximum and minimum errors between predicted and target values are relatively small. This indicates the model's high accuracy in forecasting the power load task.

Additionally, to investigate the performance at different time steps, experiments are specifically conducted on the T-Transformer module. We record the results at different time steps to observe the sensitivity of LoadSeer. The 140th to 210th time steps are shown in Figure 5. The vertical axis refers to the load ratio, which is the ratio of the actual load value to the full load of 4100. The blue line represents the

TABLE 1. Experimental Results on the PeMSD Dataset. Each algorithm’s four values on each indicator represent the calculation results from 15 minute prediction to 60 minute prediction.

Model	PeMSD4 (15/30/45/60min)		
	MAE	MAPE(%)	RMSE
ARIMA	2.51/ 2.75/ 2.96/ 3.21	5.32/ 5.69/ 6.01/ 6.56	5.72/ 6.34/ 6.85/ 7.36
LSGCN	1.45/ 1.82/ 2.04/ 2.22	2.90 / 3.84/ 4.42/ 4.85	2.93/ 3.92/ 4.47/ 4.83
AGCRN	3.32/ 3.47 /3.91 /4.07	4.17 /4.52 /4.65 /4.86	4.65/ 4.79 /4.98/ 5.12
ASTGNN	2.24/ 2.51/ 2.74/ 3.75	4.92/ 5.13/ 5.37/ 5.60	2.79/ 3.54/ 4.72/ 4.98
LoadSeer	1.32/ 1.47/ 1.90/ 2.10	3.12/ 3.47/ 4.10/ 4.71	2.07/ 2.35/ 2.67/ 2.91

Model	PeMSD7 (15/30/45/60min)		
	MAE	MAPE(%)	RMSE
ARIMA	5.57/ 5.94/ 6.27/ 6.68	13.04/ 14.01/ 15.01/ 16.78	9.00/ 9.22/ 9.43/ 9.68
LSGCN	2.22 / 2.96/ 3.43/ 3.81	5.14/ 7.18/ 8.51/ 9.60	3.98 / 5.47/ 6.39/ 7.09
AGCRN	3.37/ 3.62/ 4.37/ 4.80	6.72/ 7.90/ 8.81/ 9.94	5.02/ 6.31/ 7.04/ 8.10
ASTGNN	3.41/ 3.80/ 4.21/ 4.72	5.97/ 7.65/ 8.98/ 9.79	4.92/ 5.73/ 6.21/ 7.12
LoadSeer	2.32/ 2.71/ 3.29/ 4.01	4.90/ 6.72/ 7.93/ 9.14	4.10/ 5.32/ 6.21/ 6.93

Model	PeMSD8 (15/30/45/60min)		
	MAE	MAPE(%)	RMSE
ARIMA	1.90/ 2.12/ 2.43/ 2.79	5.11/ 5.21/ 5.46/ 5.62	4.87/ 5.24/ 5.63/ 6.22
LSGCN	1.16/ 1.46/ 1.66/ 1.81	2.24/ 3.02/ 3.51/ 3.89	2.45/ 3.28/ 3.75/ 4.11
AGCRN	1.07 / 1.32/ 1.73/ 1.90	2.10/ 2.93/ 3.32/ 3.91	2.79/ 3.84/ 4.52/ 5.31
ASTGNN	1.53/ 1.78/ 2.01/ 2.21	2.41/ 3.25/ 4.10/ 4.83	2.53/ 3.45/ 4.31/ 5.28
LoadSeer	1.13/ 1.28/ 1.40/ 1.62	2.03/ 2.71/ 3.20/ 3.71	2.19/ 3.11/ 3.63/ 4.02

TABLE 2. Results on the Flowload dataset.

Model	RMSE	MAE	MAPE
LoadSeer	38.18	31.09	14.46
LSGCN	57.06	34.14	25.64
AGCRN	45.31	26.21	16.61
ASTGNN	40.52	23.83	14.45

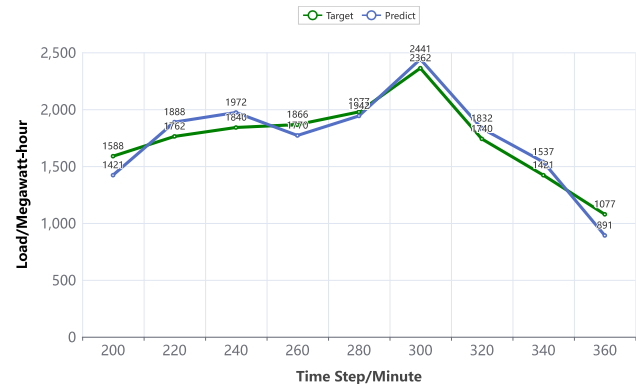


FIGURE 4. Comparison of predicted values and actual values.

ratio of the actual load value during training to full load (4100). The purple line shows the ratio of the actual load value during testing to full load (4100). We use the data for training before time step 186, and for testing after time step 186. The green line represents the ratio of the predicted load value to the full load (4100).

The load ratio from 260 to 330 time steps is shown in Figure 6. Training occurs from time step 260 to 306, and testing occurs from time step 306 to 330. It is clear that our model exhibits higher prediction accuracy at smaller time steps, and this accuracy gradually decreases with increasing time steps. This decline is attributed to the fact that the tensor-based transformer focuses more on recent times. Therefore, better prediction results are obtained in shorter time steps.

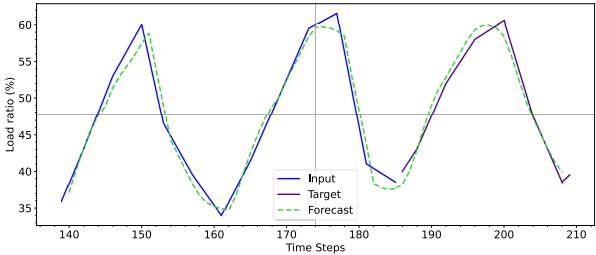


FIGURE 5. Comparison of load ratio from 140 to 210 time Steps.

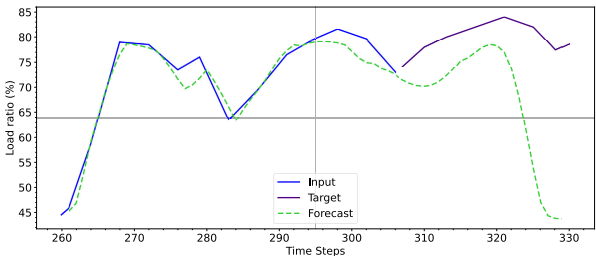


FIGURE 6. Comparison of load ratio from 260 to 330 time Steps.

The load ratio from 140 to 210 time steps is shown in Figure 6. As in Figure 5, training occurs from time step 140 to 190, and testing occurs from time step 190 to 210. It is clear that our model exhibits higher prediction accuracy at smaller time steps, and this accuracy gradually decreases with increasing time steps. This decline is attributed to the fact that the tensor-based transformer focuses more on recent times. Therefore, better prediction results are obtained in shorter time steps.

F. ABLATION STUDY

To verify the necessity of the various components of LoadSeer, we launch ablation experiments on each of them.

1) ABLATION EXPERIMENTS ON SPATIAL DATA CORRELATION USING GCN

Unlike traditional time-series forecasting models, we incorporate spatial relationship information to more accurately predict load. By converting spatial information represented by distance into an adjacency matrix, high-dimensional and complex spatial information can be quantitatively represented. Then, the complex spatial information is refined using graph-based structure modeling and by combining the advantages of GCN in associating large amounts of data. To verify the superiority of this new structure, we conduct ablation experiments on GCN. Without GCN, we compress spatial information into the same dimension as temporal information for fusion and perform the ablation experiments on the PeMSD8 dataset. The results are shown in Table 3. It can be seen that in the absence of GCN, the model cannot maintain low losses in RMSE, MAE, and MAPE. The existence of GCN improves the utilization of the spatial data in the model, thereby significantly improving the accuracy of prediction.

TABLE 3. GCN ablation experimental results.

LoadSeer		PeMSD8		
GCN	T-Transformer	RMSE	MAE	MAPE
-	✓	4.51	3.77	3.12
✓	✓	3.32	2.45	2.32

2) FOR ABLATION EXPERIMENTS WITH THE T-TRANSFORMER

To verify that our designed T-Transformer has excellent performance, we conduct ablation experiments on it. For comparison, we choose a universal Transformer architecture as a substitute. We conduct experiments on the FlowLoad dataset, which is characterized by multi-dimensional and long sequences. The dataset can effectively test the model's ability to capture and remember information. As shown in Table 4, our designed T-Transformer has a significant advantage over the traditional Transformer on long-sequence data. By slicing and finely modeling the sequences, we can avoid the problem of being unable to capture distant sequences.

In order to observe the utility of the two different Transformers in more detail, we conduct experiments on the T-Transformer and the traditional Transformer for the first 100 time steps. As shown in Figure 7, with the horizontal axis representing the time steps and the vertical axis representing the ratio of predicted values to true values, we conduct several experiments on the two models separately to avoid randomness. The randomness is represented by shaded lines in the figure. We select a representative experiment for marking.

As shown in the figure, our T-Transformer maintains superior accuracy within 100 time steps. Its accuracy decreases slowly, reaching about 65 at 100 time steps. Although the traditional Transformer has accuracy similar to that of the T-Transformer at short time steps, it decreases rapidly

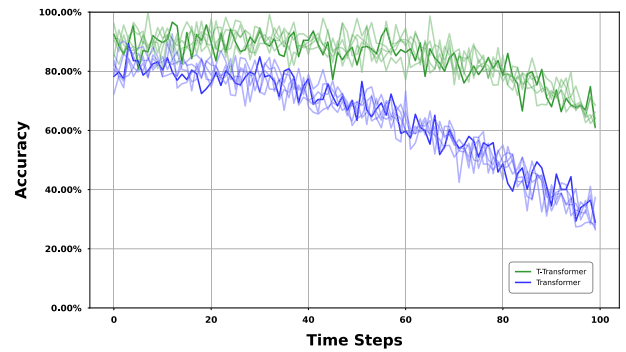


FIGURE 7. Comparative accuracy of two different transformers for 100 time steps on the FlowLoad dataset.

TABLE 4. The ablation experimental results of T-Transformer.

LoadSeer		FlowLoad		
T-Transformer	Transformer	RMSE	MAE	MAPE
-	✓	42.13	33.21	17.92
✓	-	38.18	31.09	14.46

as the time step increases, dropping to only 30 at 100 time steps. This also proves the ability of the T-Transformer to capture and remember information over long distances.

V. CONCLUSION

In response to the issue of diverse influencing factors in spatio-temporal prediction, a multidimensional spatio-temporal forecasting model, LoadSeer, is proposed by integrating tensor time series with graph convolutional neural networks. We synthesize information from multiple dimensions such as time, space, and environment and for the first time, we take into account the flow of human beings as an influencing factor, and prove its effectiveness. The efficient combination of these dimensions, together with our proposed LoadSeer to accurately capture the impact of multi-dimensional information on electricity load at each time period, is demonstrated in multiple datasets and real-world scenarios of the Asian Games.

VI. FUTURE WORK

Although LoadSeer effectively balances power load forecasting accuracy, there are still areas for future exploration. The scalability of the model could be improved by considering distributed or parallel computing techniques, especially when the number of nodes in the graph increases. Moreover, addressing the decline in accuracy for long forecasting horizons with multi-scale temporal models could further enhance its performance in real-world applications. In the future, we will focus on the above topics for further study.

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